

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

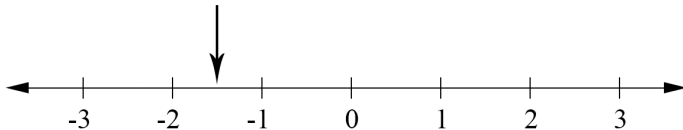
1. Number and Algebra, NAP-14484

$$5 \times ? + 15 = 95$$

What value would make this number sentence correct?

- 10 14 16 20
☐ ☐ ☐ ☐

2. Number and Algebra, NAP-85051



The arrow points to a position on the number line.

What number is at this position?

3. Number and Algebra, NAP-32091

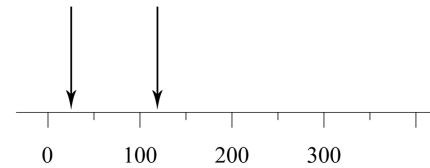
$$2.5 \times A = 0.5$$

Find the value of A that makes this number sentence correct.

- 1.25 -2.0 0.2 5.0
☐ ☐ ☐ ☐

4. Number and Algebra, NAP-49700

Two arrows point to numbers on a number line.

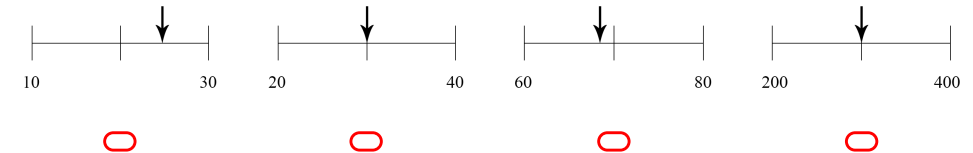


Which one of these numbers lies between the two arrows?

- ☐ 10
☐ 30
☐ 125
☐ 250

5. Number and Algebra, NAP-26180

Which arrow is pointing to the approximate position of $\sqrt{600}$ on a number line?



6. Number and Algebra, NAP-96741

Tran is a baker and sells 100 loaves of bread at the markets. Each loaf costs him \$2.25 to make.

He sells 80 loaves for \$4.95 each.

In the last hour of the market, he sells the last 20 loaves for \$2 each.

What is the total profit or loss of the 100 loaves?

- \$5.00 \$211.00 \$216.00 \$436.00
☐ ☐ ☐ ☐

7. Number and Algebra, NAP-55645

a is a positive whole number.

Which of the equations below is not true?

☐ $a + a + a = 3a$

☐ $a^3 = 3 \times a$

☐ $-a + a = 0$

☐ $a \div a = 1$

8. Number and Algebra, NAP-96774

The equation $3y - 1 = x$ is rearranged so that y is the subject.

Which of these correctly gives y as the subject?

$y = \frac{x-1}{3}$

$y = \frac{x+1}{3}$

$y = 3x - 1$

$y = 3x + 1$



Worked Solutions

1. Number and Algebra, NAP-14484

$$5 \times ? + 15 = 95$$

$$5 \times ? = 80$$

$$\therefore ? = \frac{80}{5}$$

$$= 16$$

2. Number and Algebra, NAP-85051

$$-1.5$$

3. Number and Algebra, NAP-32091

$$2.5 \times A = 0.5$$

$$\therefore A = \frac{0.5}{2.5}$$

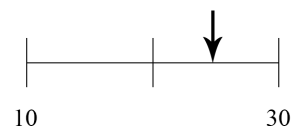
$$= 0.2$$

4. Number and Algebra, NAP-49700

$$30$$

5. Number and Algebra, NAP-26180

$$\sqrt{600} = 24.49...$$



6. Number and Algebra, NAP-96741

$$\begin{aligned}\text{Total profit/loss} &= \text{sales revenue} - \text{costs} \\ &= (80 \times 4.95 + 20 \times 2) - (100 \times 2.25) \\ &= 436 - 225 \\ &= \$211\end{aligned}$$

7. Number and Algebra, NAP-55645

$$\begin{aligned}a^3 &= a \times a \times a \\ &\neq 3 \times a\end{aligned}$$

8. Number and Algebra, NAP-96774

$$\begin{aligned}3y - 1 &= x \\ 3y &= x + 1 \\ y &= \frac{x + 1}{3}\end{aligned}$$